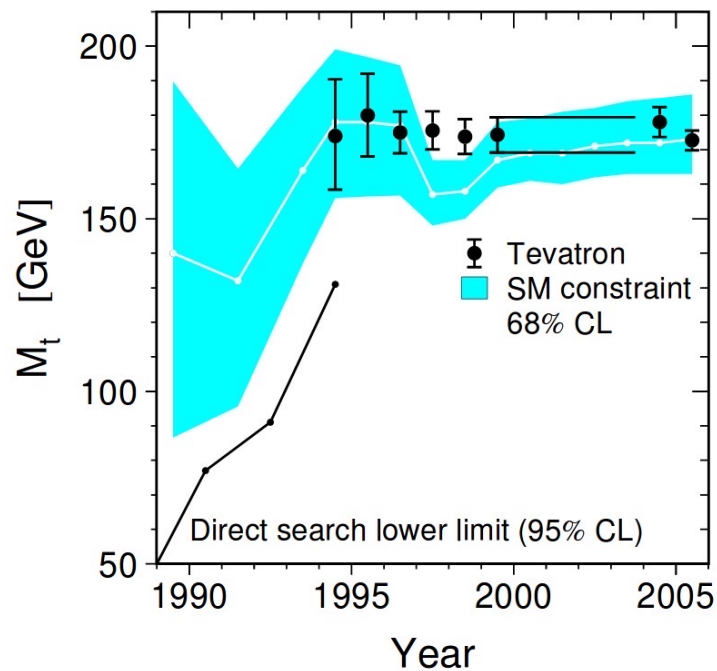


Top Quark

LEP: e^+e^- @ $\sqrt{s} = 90 \text{ GeV} \rightarrow 202 \text{ GeV}$

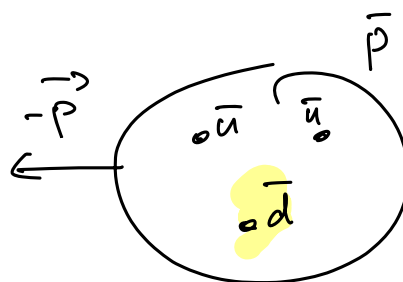
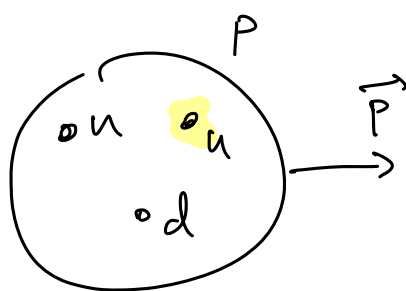
no direct sign of top production



Tevatron @ Fermilab

$p + \bar{p} \rightarrow \text{hadrons}$

$\sqrt{s} = 1.96 \text{ TeV} \Rightarrow E_{\text{beam}} \approx 1 \text{ TeV}$



$$\underline{P}_P = (E, \vec{P})$$

$$\underline{P}_{\bar{P}} = (E, -\vec{P})$$

$\sqrt{s}$: total energy in $p-\bar{p}$ collisions

$$\underline{P}_1 = x_1 \underline{P}_P$$

x_1 : x of parton/quark in proton

$$\underline{P}_2 = x_2 \underline{P}_{\bar{P}}$$

x_2 : x of quark in $\bar{p}$

$E \gg m_p \Rightarrow p, \bar{p}$ are massless

parton-parton total energy

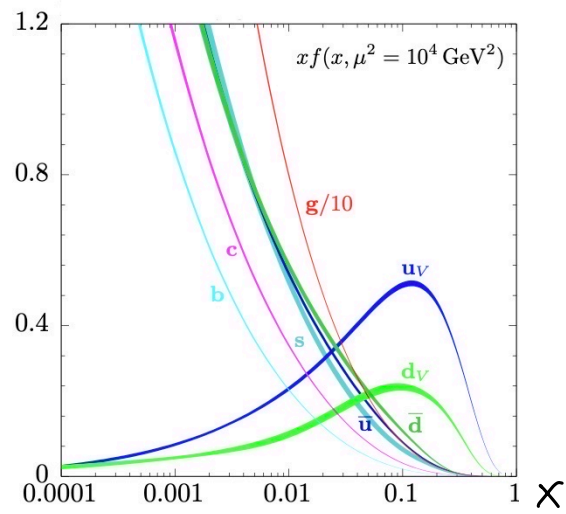
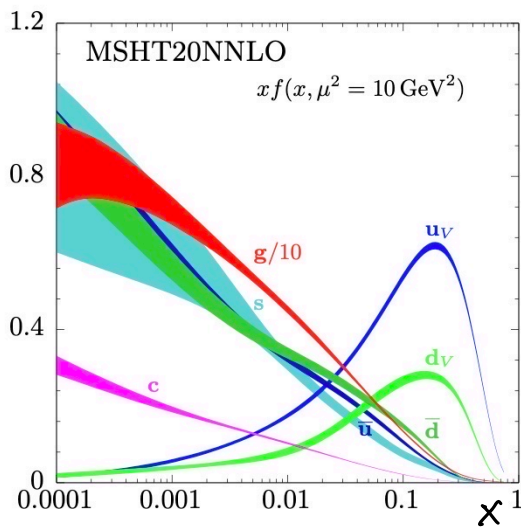
$$S_{pp} = (\underline{p}_1 + \underline{p}_2)^2 = ((x_1 + x_2) E)^2 - [(x_1 - x_2) \vec{p}]^2$$

assume $x_1 \approx x_2 \approx \langle x \rangle$ $\langle x \rangle \approx \frac{1}{3}$

$$\Rightarrow \sqrt{S_{pp}} = (x_1 + x_2) E \approx 2 \langle x \rangle E = \langle x \rangle \sqrt{S}$$

1.96 TeV

$$\sqrt{S_{pp}} \approx \frac{1}{3} \sqrt{S} = \frac{1}{3} 1.96 \approx 670 \text{ GeV}$$



1) $\sqrt{S_{pp}} < \sqrt{S}$ (proton-anti proton)

2) Lorentz boost varies for each collision
because x_1, x_2 randomly extracted from pdf
 $z \longrightarrow$

$x_1 \neq x_2$

$\xrightarrow{x_1 E} \quad \xleftarrow{x_2 E}$

$x_2 > x_1$

$\vec{P}_{in} = (x_1 - x_2) \vec{p}$

$\longleftarrow$ boost. along $-z$

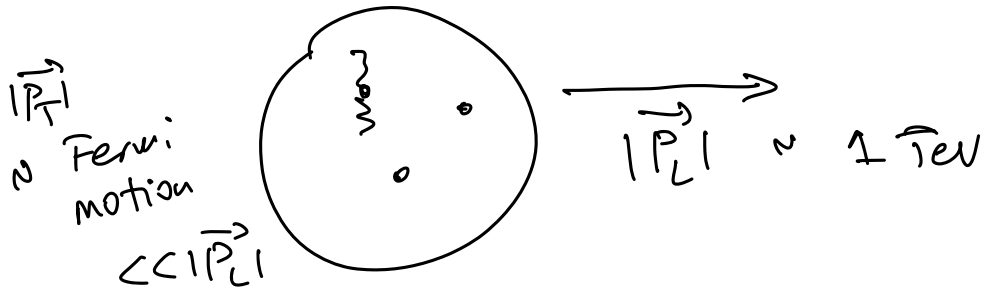
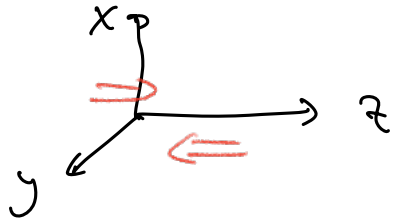
if $x_1 > x_2$

$\xrightarrow{x_1 E} \quad \xleftarrow{x_2 E}$

$x_1 - x_2 > 0 \Rightarrow$ boost along $+z$

3) Lorentz boost of center-of-mass of parton collision
is unknown.

Lorentz boost always along z axis.



Initial state:

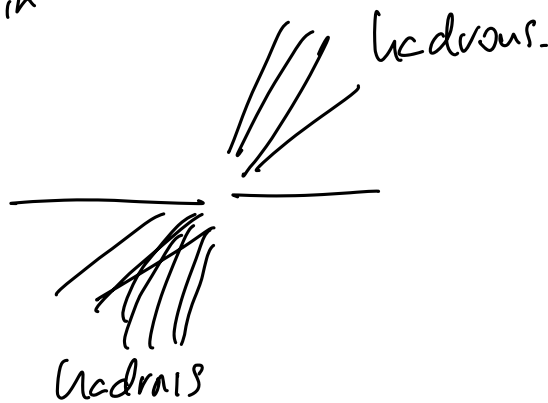
$$\vec{P}_T = \vec{0}$$

$$P \longrightarrow \longleftarrow \bar{P}$$

before the collision. $\vec{P}_L = \vec{0}$
for quarks/leptons $\vec{P}_{L1} + \vec{P}_{L2} \neq \vec{0}$

Final state

$$\vec{P}_{T \text{ fin}} = \vec{0} \quad \text{because boost only along } \pm z$$



Final state $\sum_i \vec{P}_{Ti} = \vec{0}$

Top Production in $P\bar{P}$

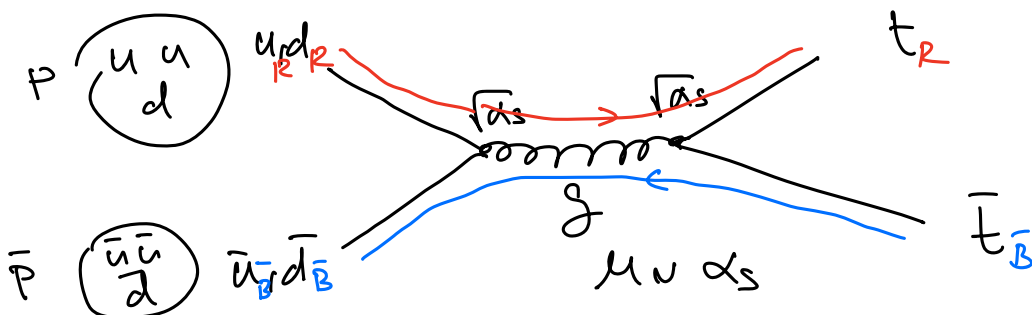
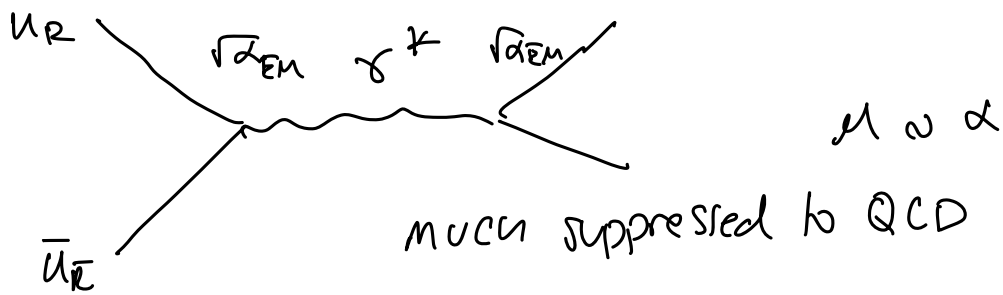


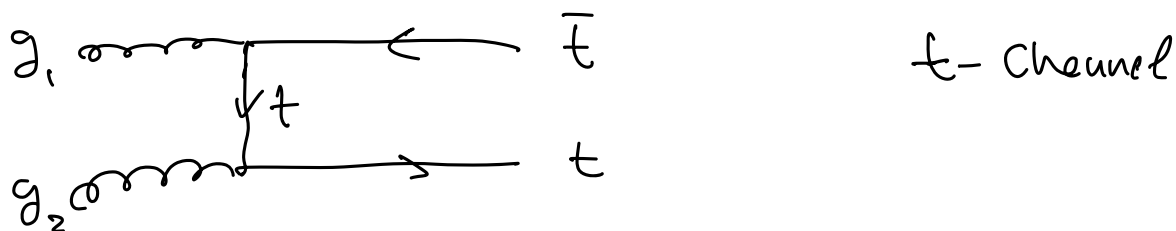
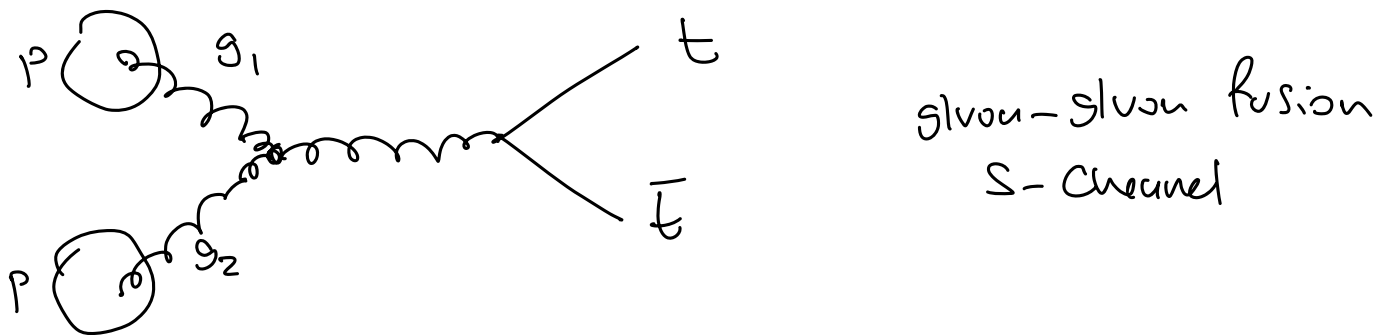
Diagram with valence quarks

$$\alpha_s = \frac{g_s^2}{4\pi}$$



S-channel dominant @ Tevatron.

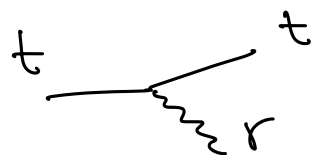
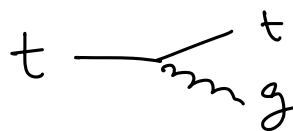
LHC : $\sqrt{s} = 13.6 \text{ TeV} \Rightarrow$ access smaller x
 $\Rightarrow$ gluons become dominant



$$m_t = 176 \text{ GeV} \quad \Gamma_t = 2 \text{ GeV}$$

$t\bar{t}$ resonance or toponium?

top can decay hadronically



weak decay $t \rightarrow b \ell \nu$

t b ℓ ν

$$\Gamma(t \rightarrow qb) \propto \left(\frac{g_W^2}{m_W^2} \right)^2 m_t^3$$

$$G_F^2$$

$$G_F \approx 1 \times 10^{-5} \text{ GeV}^{-2}$$

$$\Rightarrow \tau_t = \frac{1}{\Gamma_t} \approx 10^{-25} \text{ sec}$$

For the longest time assumption: top too heavy to form $(t\bar{t})$ state before decay.

$$t\bar{t} \text{ threshold} = 2m_t \approx 350 \text{ GeV}$$

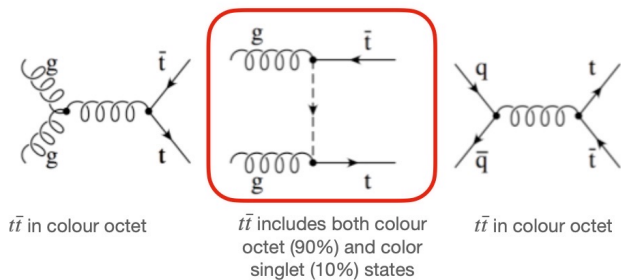
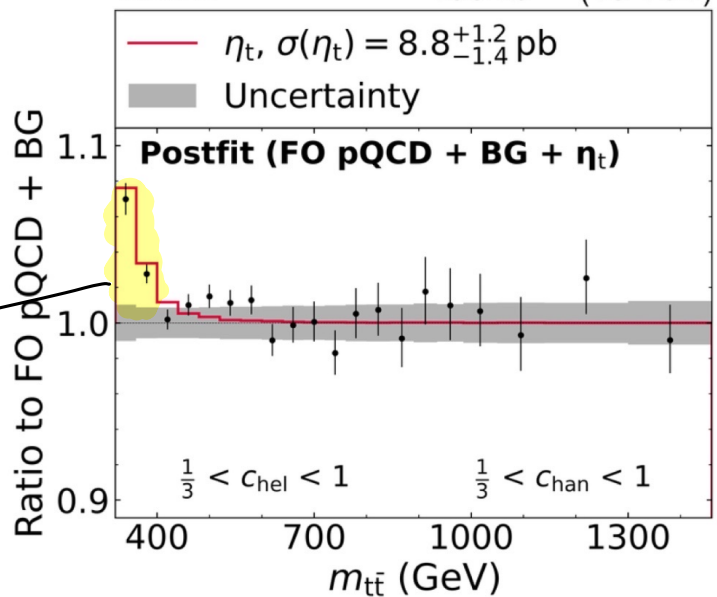
FO pQCD:

Fixed-order perturbative QCD

toponium?

CMS

138 fb⁻¹ (13 TeV)

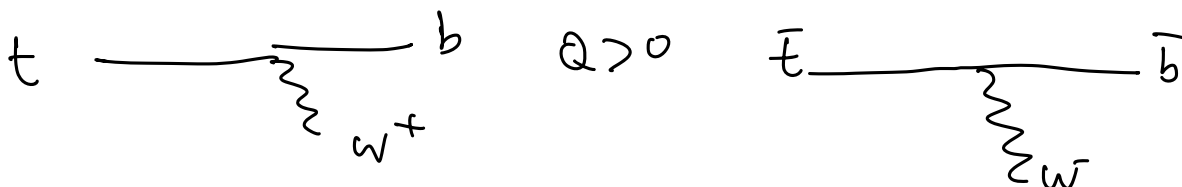


$t\bar{t}$ color singlet

$$1s^0 \equiv \eta_t$$

No toponium at Tevatron:

$$p + \bar{p} \rightarrow t + \bar{t} + (X)$$



$$m_t = 170 \text{ GeV}$$

$$m_W = 80 \text{ GeV}$$

$$m_b \approx 5 \text{ GeV}$$

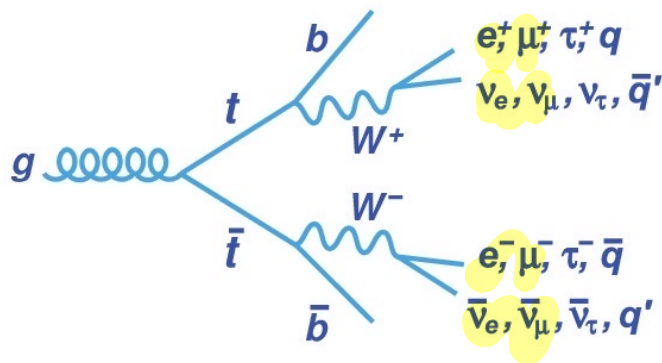
$$p + \bar{p} \longrightarrow t + \bar{t} \begin{cases} \longrightarrow b + W^- \\ \longrightarrow b + W^+ \end{cases}$$

$$W^- \begin{cases} \longrightarrow \ell^-, q_1, u \\ \longrightarrow \bar{\nu}_\ell, \bar{q}_2, \bar{d} \end{cases}$$

W [±] DECAY MODES	Fraction (Γ_i/Γ)	Confidence level	p (MeV/c)
$\ell^+ \nu$	[b] (10.86 ± 0.09) %	—	—
$e^+ \nu$	(10.71 ± 0.16) %	40189	—
$\mu^+ \nu$	(10.63 ± 0.15) %	40189	—
$\tau^+ \nu$	(11.38 ± 0.21) %	40170	—
hadrons	(67.41 ± 0.27) %	—	—

$\ell: e, \mu, \tau$

to have the best mass resolution $\Rightarrow$ smallest # of hadronic jets



golden channel:
 2 b-quark jets
 2 leptons of opposite charge.
 2 neutrinos
 (missing energy)